Mini Assessment

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Task	Mini Assessment

1. If an API returns 429, what should your program do?

• 429 means “Too Many Requests.”

• It usually happens when too many requests are sent within a short period.

• The program should wait before trying again instead of immediately sending another request.

• A retry with a delay or backoff is a good approach.

• If the API gives a Retry-After value, the program can use it.

• This helps avoid unnecessary requests and reduces the load on the server.

Example: If the API returns 429, the program can wait for 2 seconds and try again. If it happens again, it can wait longer before the next attempt.

2. If response.json() fails, what should your program do?

• response.json() is used when we expect the API to return JSON data.

• Sometimes the server may return plain text, an HTML error page, an empty response, or invalid JSON.

• In that situation, response.json() can raise an error.

• The program should handle the error instead of crashing.

• Using try/except allows the program to handle the problem safely.

• Checking the HTTP status code and response text can also help identify what went wrong.

Example: If a server returns an HTML error page instead of JSON, try/except can catch the problem and display a useful message instead of stopping the program.

3. If a JSON key is missing, why might .get() be safer than []?

• API responses do not always contain every expected field.

• Using data["name"] assumes that the "name" key exists.

• If the key is missing, Python raises a KeyError.

• data.get("name") is safer because it returns None when the key is missing.

• A default value can also be provided with .get().

• This makes the program more reliable when working with real-world API data.

Example: data.get("name", "Unknown") returns "Unknown" if the "name" key is missing, so the program can continue normally.